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4. A Welsh farmer decides to purchase a wind generator to provide some electrical energy for his farm.
A selection of wind generators is available.

	Wind generator 1	Wind generator 2	Wind generator 3
Cost to buy and install (£)	1800	3000	2925
Expected lifetime (years)	15	15	15
Estimated number of units produced per year (kWh)	2000	4000	3000
Cost per unit (£)	0.15	0.15	0.15
Estimated saving per year (£)		600	450
Estimated payback time (years)		5.0	6.5

(a) (i) Use the equation:

$$\text{estimated saving per year} = \text{units produced per year} \times \text{cost per unit}$$

and data from the table to calculate the estimated saving per year if **wind generator 1** was installed. [1]

Estimated saving per year = £ ....

(ii) Use the equation:

$$\text{estimated payback time} = \frac{\text{cost to buy and install}}{\text{estimated saving per year}}$$

and data from the table to calculate the estimated payback time for **wind generator 1**. [2]

Estimated payback time = ... years

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(b) (i) A salesperson suggests to the farmer that **wind generator 3** is better than **wind generator 2** because it is cheaper.
Explain, using information from the table, if the salesperson's suggestion is correct over a 5-year timescale. [2]

....

....

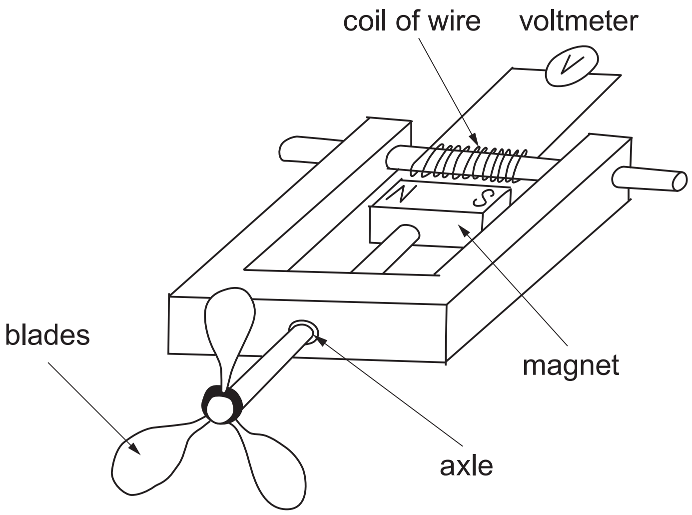
....

(ii) Give **two** reasons why the payback time can only be an estimate. [2]

1. ....

2. ....

(c) A wind generator only operates in certain wind speeds.
The diagram below shows a student's design for measuring wind speed.



(i) When the wind blows a reading on the voltmeter is obtained.
The following sentences describe the different stages.

A The magnetic field lines cut through the coil of wire.
B The blades rotate.
C The magnet rotates.
D This induces a voltage in the coil of wire.

Put the letters **A**, **B** and **C** in the boxes to correctly describe the process of inducing a voltage in the coil of wire. The final box has already been completed, [2]

	→		→		→	**D**

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© WJEC CBAC Ltd. (3420U10-1) **Turn over.**

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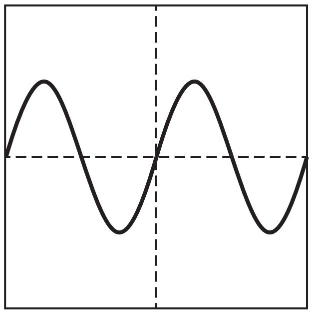
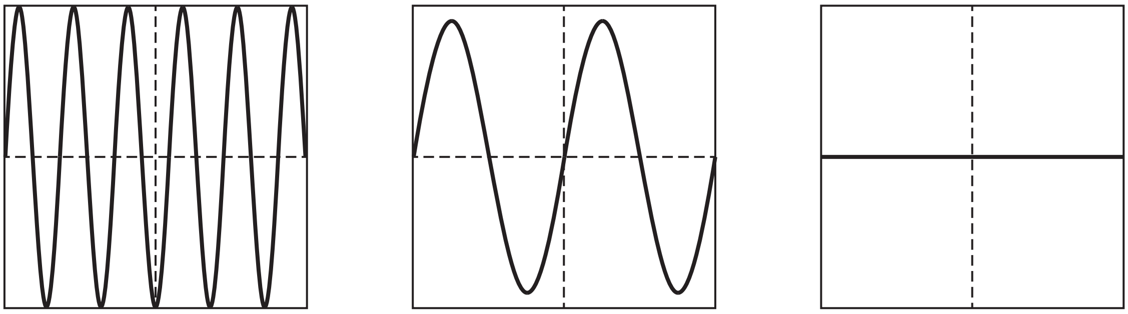
(ii) When the wind speed is **2 m/s** the alternating voltage produced is shown below.



The following traces **X**, **Y** and **Z** are produced during different wind conditions.



Trace **X** Trace **Y** Trace **Z**

I. State which trace (**X**, **Y** or **Z**) is produced when the wind speed is **faster than 2 m/s**.

Trace .... [1]

II. State which trace (**X**, **Y** or **Z**) is produced when **the wind stops**.

Trace .... [1]

(d) When the student's design is built it is found that the voltmeter is not sensitive enough to measure a wind speed less than 0.5 m/s.

(i) Suggest **one** change that can be made to the **magnet** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

....

(ii) Suggest **one** change that can be made to the **coil of wire** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

....

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